

# MIKHERB AI

AI-Powered Herbicide Discovery & Formulation Intelligence

## Complete Product Requirements Document (PRD)

**Organization:** Miklens Bio Pvt. Ltd.

**Version:** 1.0   **Purpose:** Antigravity / AI-agent implementation

**Local-first • Open-source • Reproducible • Automated • Research-focused**

Important: This platform generates and prioritizes scientific hypotheses. Computational predictions are not proof of herbicidal efficacy, safety, regulatory compliance, or freedom-to-operate.

# 1. Product Vision

MikHerb AI should function as a virtual herbicide R&D scientist. A researcher should be able to define a weed, crop and research objective and have the system automatically execute target discovery, protein analysis, crop-selectivity analysis, chemical-library construction, AI screening, docking, physicochemical filtering, safety/environmental screening, candidate ranking and report generation.

```
WEED → Biology → Target → Protein → Crop selectivity → Structure
→ Binding pocket → Chemical library → RDKit → GNINA → Boltz-2
→ Consensus ranking → Safety/novelty → Optional MD → Experimental shortlist
→ Miklens experiments → QSAR/active learning → Next discovery cycle
```

## 2. Zero-Cost / Local-First Philosophy

The software should have no mandatory paid software, SaaS subscription, proprietary AI API or commercial docking dependency. The system must remain useful with local CPU-only execution and become faster when an NVIDIA GPU is available.

- Software cost target: **■0** mandatory.
- Compute cost is hardware-dependent; the UI must clearly distinguish software cost from compute cost.
- No proprietary cloud AI API should be required for the scientific pipeline.
- After models and public datasets are downloaded, the platform should support offline operation.
- Miklens confidential data should remain local by default.

## 3. Technology Stack

| Layer             | Recommended technology                               |
|-------------------|------------------------------------------------------|
| Frontend          | React + Vite + TypeScript + Tailwind CSS + shadcn/ui |
| Backend           | Python + FastAPI + Pydantic                          |
| Database          | PostgreSQL + pgvector; SQLite fallback               |
| Scientific Python | RDKit, NumPy, Pandas, SciPy, scikit-learn, PyTorch   |
| Job system        | Redis + Celery/RQ                                    |
| Containers        | Docker / Docker Compose                              |
| 3D visualization  | 3Dmol.js / Three.js                                  |
| Reports           | Python + ReportLab                                   |
| Local LLM         | Ollama or configurable local inference server        |

## 4. Core Scientific Engine Architecture

### 4.1 Protein Intelligence

- UniProt: sequences, annotations, gene/protein identity, function and cross-references.
- AlphaFold DB: predicted protein structures when experimental structures are unavailable.
- Foldseek: rapid structural-homolog and structural-comparison analysis.
- P2Rank: binding-pocket prediction directly from protein structures.
- Protein Analyzer: sequence length, domains, motifs, catalytic residues, transmembrane regions, disorder and structure confidence.

### 4.2 Chemical Intelligence

- RDKit: SMILES validation, standardization, salt removal, descriptors, fingerprints, similarity, clustering and substructure search.
- PubChem: public compound/structure information and programmatic retrieval.
- ChEMBL: public bioactivity, assay, compound and target relationships.
- User-uploaded SDF/CSV/SMILES/MOL/MOL2 libraries.
- Chemical Library Builder: download → parse → standardize → deduplicate → descriptor calculation → filtering → clustering → database.

### 4.3 AI Structure and Binding

- Boltz-2: primary AI complex-structure and binding-affinity engine.
- GNINA: independent deep-learning docking/scoring engine.
- DiffDock: optional additional pose-generation/scoring engine.
- Consensus engine: never rely on one model; compare independent predictions and pose agreement.

### 4.4 Molecular Dynamics

- OpenMM for optional high-detail simulation of only the top candidates.
- Do not run expensive MD on the full chemical library.

## 5. End-to-End Automated Discovery Pipeline

1. Select weed and crop
2. Retrieve/curate biological information
3. Identify candidate essential targets
4. Retrieve target sequences
5. Identify crop homologs
6. Acquire or predict protein structures
7. Assess structure confidence
8. Predict binding pockets
9. Build chemical library
10. RDKit cleaning and physicochemical filtering
11. GNINA screening
12. Boltz-2 screening
13. Optional DiffDock screening
14. Consensus scoring
15. Weed-vs-crop selectivity analysis
16. Environmental/toxicity evidence screening
17. Chemical similarity/novelty analysis
18. Optional OpenMM MD for top candidates
19. Final candidate ranking
20. Add selected candidates to experimental queue
21. Import experimental results

## 6. Discovery Project Wizard

- Project creation: project name, researcher, crop, weed, objective and notes.
- Objectives: new herbicide, new target, improve selectivity, improve grass activity, improve broadleaf activity, improve perennial activity, improve translocation, improve persistence, or formulation optimization.
- One-click RUN DISCOVERY starts the resumable workflow.
- Every stage exposes status: pending, running, completed, failed, skipped.
- Failed stages can be restarted without repeating successful upstream stages.

## 7. Target Discovery Engine

The system should combine biological evidence, target essentiality, weed specificity, crop divergence, structure confidence, druggability and existing evidence into a configurable Target Opportunity Score.

| Factor               | Example |
|----------------------|---------|
| Essentiality         | 0–10    |
| Weed specificity     | 0–10    |
| Crop divergence      | 0–10    |
| Structure confidence | 0–10    |
| Druggability         | 0–10    |
| Existing evidence    | 0–10    |

## 8. Crop Selectivity Engine

Crop selectivity is a core differentiator. The platform should compare weed and crop homologs at sequence, structural, pocket and predicted-binding levels.

```
WEED TARGET
↓ sequence/structure/pocket comparison
CROP HOMOLOG
↓ dual prediction
Boltz + GNINA + structural comparison
↓
SELECTIVITY SCORE
```

- Sequence identity and residue differences.
- Binding-pocket similarity and divergent residues.
- Boltz predicted affinity comparison.
- GNINA docking comparison.
- Pose/interactions comparison.
- Configurable selectivity score with transparent evidence.

## 9. Chemical Compatibility / Formulation Intelligence

Create a separate Formulation Lab. It should record active ingredient, solvent, surfactant, adjuvant, acid/base, buffer, preservative and water components and flag predicted risks.

- pH and ionization effects.
- Solubility and precipitation risk.
- Phase separation risk.

- Potential crystallization.
- Oxidation/hydrolysis risk flags.
- Surfactant and component interaction flags.
- Estimated physicochemical compatibility.
- Every result must be labelled as a prediction requiring laboratory validation.

## 10. Consensus / MikHerb Score

Candidate ranking must not depend on a single docking score. The score should combine evidence from multiple independent sources and keep all weights configurable.

```
MikHerb Score =
Target relevance
+ Pocket confidence
+ Boltz prediction
+ Boltz confidence
+ GNINA
+ optional DiffDock
+ Pose agreement
+ Crop selectivity
+ Physicochemical suitability
+ Chemical novelty/similarity
+ Safety/environment evidence
```

The application should display component scores and the overall score. A score is a prioritization metric, not experimental proof.

## 11. Safety, Environmental and Novelty Screening

- Preliminary human-toxicity evidence.
- Aquatic toxicity evidence.
- Pollinator/bee concern where public evidence exists.
- Persistence, mobility and bioaccumulation evidence.
- Known restrictions or regulatory concerns.
- Known-herbicide similarity and Tanimoto similarity.
- Computational novelty/similarity estimate.
- Never label a candidate 'safe' or 'patent safe' without appropriate evidence/legal review.
- Unknown evidence must remain UNKNOWN rather than being converted to safe.

## 12. Experimental Learning System

Every Miklens experiment should become structured data so that the platform can learn from internal evidence.

- Compound/candidate ID.
- Target and predicted mechanism.
- Dose and application method.
- Weed/crop species and growth stage.
- Environmental conditions.
- Visual injury, biomass reduction, mortality, recovery and regrowth.
- Crop phytotoxicity.
- Replicate-level observations.
- Biochemical, pot, greenhouse and field results.

## 13. Active Learning

Once sufficient proprietary data exist, the system should learn not only which candidates have the highest predicted activity, but which experiment would provide the most useful information. Predictions must include uncertainty so scientists can choose informative experiments.

## 14. Generative Chemistry — Later Phase

Molecular generation should not be the first milestone. First establish a reliable screening and validation system. Later, add scaffold-based or graph-based generation followed by RDKit validation, physicochemical filtering, Boltz/GNINA screening, crop selectivity and safety filtering.

## 15. Literature and Research Intelligence

- Store papers with title, authors, year, DOI, target, species, chemical, result and evidence level.
- Never fabricate literature or citations.
- Scientific claims displayed in reports should link to stored evidence.
- The research agent should use tools/data rather than inventing numerical scientific results.

## 16. AI Research Agent

```
AI Agent
■■■ target_search()
■■■ protein_analysis()
■■■ structure_search()
■■■ pocket_prediction()
■■■ library_builder()
■■■ rdkit_filter()
■■■ gnina()
■■■ boltz()
■■■ diffdock()
■■■ selectivity()
■■■ toxicity()
■■■ md()
■■■ report()
```

The agent should orchestrate scientific tools, explain results and generate reports. Numerical outputs must come from scientific engines, not from language-model invention.

## 17. Main Application Workspaces

| Workspace   | Purpose                                                               |
|-------------|-----------------------------------------------------------------------|
| DISCOVERY   | Create and run herbicide discovery projects.                          |
| TARGETS     | Protein, pathway, structure, pocket and selectivity analysis.         |
| CHEMISTRY   | Chemical libraries, descriptors, similarity and screening.            |
| FORMULATION | Compatibility, stability and formulation experiments.                 |
| EXPERIMENTS | Biochemical, pot, greenhouse, RCBD and field trials.                  |
| AI LAB      | QSAR, active learning, model registry and proprietary Miklens models. |

## 18. Candidate Detail Page

```
CANDIDATE MH-000127
MikHerb Score: 91/100
Evidence Level: 1 – Computational hypothesis

Target: Target-X
Predicted affinity: High
Crop selectivity: High
```

```
Novelty: High
Physicochemical suitability: Good
Safety evidence: Unknown
```

```
3D complex viewer
```

```
Boltz ██████████
GNINA ██████████
DiffDock ██████████
Pose agreement: HIGH
```

```
[Add to Experimental Queue]
```

## 19. Experiment & Trial Integration

- Support biochemical assay, CRD, RCBD, pot, greenhouse and field trials.
- Store treatment, dose, replicate, DAT, injury %, biomass, regrowth and crop-injury measurements.
- Calculate mean, SD, SE, CV, ANOVA and treatment comparisons.
- Allow import from CSV/Excel.
- Feed validated experimental outcomes into QSAR and active-learning models.

## 20. Reproducibility & Provenance

Every run must record software/model/database versions, parameters, inputs, outputs, timestamps, hardware and random seeds. A project should be exportable as a complete reproducible research package.

```
project/
████ metadata.json
████ targets/
████ proteins/
████ compounds/
████ docking/
████ boltz/
████ md/
████ reports/
████ provenance.json
```

## 21. Dashboard Requirements

- Active projects.
- Targets analysed.
- Chemical library size.
- Compounds screened.
- High-priority candidates.
- Candidates awaiting experimental testing.
- Experimental evidence level distribution.
- Failed/running jobs.
- GPU/CPU status.
- Database/model update status.

## 22. Hardware & Execution

On startup, run a hardware check and show CPU, RAM, GPU and VRAM. Enable GPU acceleration when available and CPU fallback otherwise. Large AI screening and molecular-dynamics jobs should be asynchronous.

```
MIKHERB-AI HARDWARE CHECK
CPU: detected
RAM: detected
```

```
GPU: detected / unavailable
VRAM: detected
Boltz-2: READY / LIMITED
GNINA GPU: READY / CPU
OpenMM GPU: READY / CPU
```

## 23. Docker / Deployment

```
mikherb/
■■■ frontend
■■■ api
■■■ worker
■■■ postgres
■■■ redis
■■■ boltz
■■■ gnina
■■■ openmm
■■■ p2rank
■■■ foldseek
■■■ nginx
```

Provide Docker Compose plus Windows/Linux setup scripts and environment validation. Keep scientific engines modular so individual tools can be upgraded without rewriting the application.

## 24. Security & Confidentiality

- Default deployment should listen on localhost.
- Optional username/password and role-based access.
- Roles: Admin, Scientist, Research Assistant, Viewer.
- Separate PUBLIC DATA and MIKLENS CONFIDENTIAL DATA.
- No proprietary data should be uploaded to external AI services by default.
- Maintain audit logs for important changes.

## 25. Model Registry

- Register Boltz-2, GNINA models, DeepChem/scikit-learn models and Miklens-specific QSAR models.
- Store model version, training data description, date, validation metrics and limitations.
- Do not allow an unvalidated ML model to be presented as scientifically reliable.

## 26. Evidence-Level System

| Level | Meaning                             |
|-------|-------------------------------------|
| 0     | Pure computational hypothesis       |
| 1     | Multiple computational models agree |
| 2     | Biochemical evidence                |
| 3     | Whole-plant evidence                |
| 4     | Greenhouse evidence                 |
| 5     | Field evidence                      |

## 27. What Must Be Fully Automated

- Protein retrieval and structure acquisition.
- Structure-quality checks.

- Pocket prediction.
- Chemical download/cleaning/descriptor calculation.
- Docking and Boltz screening.
- Consensus scoring.
- Selectivity analysis.
- Report generation.
- Database updates.
- Failed-job retry and resumable pipelines.
- Backups and provenance capture.

## 28. Human Approval Gates

- Final target selection.
- Final candidate selection.
- Chemical procurement/synthesis decisions.
- Experimental design.
- Biochemical testing.
- Greenhouse and field testing.
- Regulatory interpretation.
- Patent/legal conclusions.

## 29. Development Roadmap

| Phase                     | Deliverable                                                            |
|---------------------------|------------------------------------------------------------------------|
| 1 — Foundation            | React/FastAPI/PostgreSQL, auth, projects, files, job queue.            |
| 2 — Protein Intelligence  | UniProt, AlphaFold/PDB, Foldseek, P2Rank, 3D viewer.                   |
| 3 — Chemical Intelligence | PubChem, ChEMBL, RDKit, chemical library.                              |
| 4 — AI Screening          | Boltz-2, GNINA, optional DiffDock, consensus ranking.                  |
| 5 — Selectivity           | Weed-vs-crop structural and binding comparison.                        |
| 6 — MD/Safety             | OpenMM, environmental/safety and novelty modules.                      |
| 7 — Experimental AI       | Assay/trial database, QSAR, active learning, formulation optimization. |

## 30. V1 Definition of Done

V1 is successful when a scientist can enter a weed, crop and target and click RUN DISCOVERY, after which the platform automatically produces a target report, protein structure, binding pocket, chemical library, RDKit filtering results, GNINA screening, Boltz-2 screening, crop-selectivity analysis and a ranked candidate list with downloadable provenance-rich reporting.

## 31. V2 Definition of Done

V2 adds DiffDock, OpenMM, environmental/safety evidence, chemical novelty/similarity, literature intelligence and patent-intelligence support.

## 32. V3 Definition of Done

V3 becomes the Miklens proprietary discovery platform by incorporating internal assay, greenhouse, field and formulation data into QSAR, uncertainty estimation, active learning and formulation optimization.

## 33. Recommended Product Name

### MIKHERB AI

AI-Powered Herbicide Discovery & Formulation Intelligence

## 34. Master Implementation Prompt for Antigravity

Use the following as the master instruction for the coding agent. Implement sequentially; do not attempt all advanced capabilities in a single generation.

Build a production-quality application called MIKHERB AI – AI-Powered Herbicide Discovery & Formulation Intelligence.

Build it as a local-first, zero-mandatory-cost scientific research platform. Do not use paid APIs, paid SaaS, proprietary cloud AI APIs, or commercial molecular-docking software as mandatory dependencies.

Use React + Vite + TypeScript + Tailwind for the frontend and Python + FastAPI for the backend. Use PostgreSQL/pgvector for production and SQLite fallback for local development. Use Redis + Celery/RQ for long-running scientific jobs.

Integrate RDKit, PubChem, ChEMBL, UniProt, AlphaFold DB, Foldseek, P2Rank, Boltz-2, GNINA, DiffDock and OpenMM through modular adapters. Every scientific engine must be independently replaceable and have its own version/configuration record.

Build six main workspaces: Discovery, Targets, Chemistry, Formulation, Experiments and AI Lab.

Primary workflow:

Weed + Crop → target discovery → protein retrieval → structure acquisition → structure quality analysis → crop homolog analysis → pocket prediction → chemical-library acquisition → RDKit filtering → GNINA screening → Boltz-2 screening → optional DiffDock → consensus scoring → crop selectivity → physicochemical filtering → environmental/toxicity screening → novelty/similarity analysis → optional OpenMM → final candidate ranking → experimental queue → experimental data → QSAR/active learning.

Implement a resumable asynchronous pipeline. Every stage must support pending/running/completed/failed/skipped states and restart without rerunning successful upstream stages.

Implement a provenance system recording input files, model versions, database versions, software versions, parameters, random seeds, timestamps, hardware and outputs.

Never fabricate scientific results. Never treat docking or AI predictions as experimental proof. Every candidate must have an evidence level from computational hypothesis through field evidence.

Build an interactive 3D protein/ligand viewer using 3Dmol.js.

Build a configurable MikHerb Score combining target relevance, pocket confidence, Boltz prediction, docking, pose agreement, crop selectivity, physicochemical suitability, novelty and safety evidence. Make weights configurable.

Build a crop-selectivity engine comparing weed and crop homologs at sequence, structural, pocket and predicted-binding levels.

Build a formulation compatibility engine for pH, ionization, solubility, precipitation, phase separation and stability-risk prediction. Clearly label all such outputs as predictions requiring laboratory validation.

Build an experimental database supporting biochemical assays, pot trials, greenhouse trials, RCBD trials and field trials. Allow validated experimental results to feed QSAR and active-learning models.

Build an AI research agent that executes scientific tools but cannot invent numerical results.

Keep Miklens confidential data local by default and separate public data from proprietary data.

Build automatic R&D report generation with target analysis, protein analysis, chemical screening, docking, Boltz, selectivity, safety, candidate ranking, experimental recommendations, limitations and provenance.

Include Docker configuration, Windows/Linux setup scripts, environment checks, GPU detection, CPU fallback, database migrations, tests, logging, backups and job recovery.

Implement in phases:  
Phase 1 Foundation

Phase 2 Protein Intelligence  
Phase 3 Chemical Intelligence  
Phase 4 AI Screening  
Phase 5 Selectivity  
Phase 6 MD/Safety  
Phase 7 Experimental AI.

Do not skip testing between phases.

## 35. Scientific / Engineering Guardrails

- Predictions are hypotheses, not proof.
- No fabricated literature, assay results or safety conclusions.
- No automatic regulatory or patent conclusions.
- All external-data imports should preserve source and retrieval metadata.
- All model outputs should preserve model/version/configuration.
- Human approval should be required before experimental actions.
- Use appropriate laboratory safety, environmental containment and legal/regulatory controls for any wet-lab work.

## 36. Reference Open-Source / Public Ecosystem

Recommended foundation components include Boltz-2, RDKit, GNINA, P2Rank, Foldseek and OpenMM, plus public resources such as AlphaFold DB, UniProt, PubChem and ChEMBL. Verify the current license, model-weight terms, database terms and redistribution conditions before packaging third-party components into a commercial distribution.